Take-away Sheets

Applied AI and Automation for Executives. One page for each session: what you keep.

Session 1

Seeing: pixels, features and neurons

How can a machine read a picture when all it has is numbers?

- SEEING: pixels, features and neurons
- A grey picture is a grid of numbers: 8 x 8 = 64. A colour pixel is 3 numbers (red, green, blue): 8 x 8 x 3 = 192.
- A feature is a measurement from the picture, such as the amount of ink in the centre.
- A neuron is a weighted checklist: tick (1 or 0) x weight, add the points, compare with a cut-off.
- Payment-risk neuron: big amount x 3, new payee x 2, after hours x 1. Flag at 4 or more.
- A network stacks layers: 64 numbers in, 16 neurons, 10 scores out. 1,210 settings, called parameters.
- Training is the loop: guess, measure the error, adjust, repeat. In the recorded run, 17% right after 1 pass and 97% after 200.
- Recorded run, 5 November 2026: 436 of 450 unseen digits right. Shift each digit by 1 pixel: 46% right.
- Ask of any image system: what was it trained on, what was it tested on, and what does a person check?

I keep: A one-page sheet: how a picture becomes numbers, the payment-risk neuron worked by hand, and the digit results.

Session 2

Learning from examples: types of machine learning, decision trees, nearest neighbours

How does a machine learn a rule from examples, and how much can I trust what it learned?

- LEARNING FROM EXAMPLES
- Four types: supervised (examples with answers), unsupervised (finds groups), reinforcement (trial and reward), self-supervised (predicts a hidden part of its own data).
- Classification picks a category. Regression predicts a number.
- A decision tree is yes-or-no questions. It reads as plain rules you can audit. A tree grown too deep memorises: 100% on old claims, 88% on new, against 93% for a small tree.
- Nearest neighbours answers from the most similar past cases: a vote of the 5 nearest. Put measurements on one scale first: 69% raw, 90% scaled.
- Cross-validation: take turns to hide part of the data. Report the average and the spread.
- Recorded run, 5 November 2026: on 40 customer messages the learned models scored about 55% (range of single tests very wide). A simple keyword rule got 23 of 40. Forty examples is too few for text.
- The expense claims are fictional. The 40 messages are Nile Valley Foods practice files.
- Ask of any model: how many examples, how was it tested, and who checks the doubtful cases?

I keep: A one-page sheet: the four types of learning, a tree read as rules, nearest neighbours, and the honest scores from today's run.

Session 3

Measuring a model: accuracy, precision, recall, F1, ROC and AUC, thresholds, baselines, overfitting

How do I know whether a model is any good, and what should I ask the vendor who says it is?

- VENDOR QUESTION SHEET: THE NUMBERS BEHIND AN AI CLAIM
- 1. What data was the figure measured on, and had the model seen that data?
- 2. How common is the thing we look for? What does always saying no score?
- 3. What are precision and recall, and at which threshold?
- 4. Who chose the threshold, and what did each kind of mistake cost?
- 5. How does it compare with a simple rule or with our current process?
- 6. How many real cases were in the test? Few cases means a shaky figure.
- 7. Can we test it on 50 of our own cases before we sign?
- 8. How will you check it after launch, as our data changes?
- 9. Who reads the false alarms, and how long does it take them?
- 10. What happens to a case the model gets wrong? Who can overrule it?
- Remember: accuracy rewards doing nothing when the thing is rare. A perfect score on old data can mean memorising.

I keep: The vendor question sheet: ten questions about the numbers behind any AI claim. Keep it next to every proposal you read.

Session 4

Words as numbers: bag of words, TF-IDF, embeddings and distance

How can a computer find the right clause or policy when I describe it in plain words?

- DISTANCE CHEAT SHEET: HOW CLOSE ARE TWO ARROWS?
- STRAIGHT LINE (Euclidean distance). Picture: a crow flying. Rule: smaller is closer. Note: includes length.
- CITY BLOCKS (Manhattan distance). Picture: a rider on the streets. Rule: smaller is closer. Note: adds up each difference.
- AGREEMENT SCORE (dot product). Picture: shopping list against stock. Rule: bigger is closer. Note: long arrows score higher.
- ANGLE (cosine similarity). Picture: two cyclists facing the same way. Rule: bigger is closer, 1 is the same way. Note: ignores length.
- Worked example: a = (3, 4), b = (6, 8), c = (4, 3). Straight line a to b 5, a to c 1.4. Blocks 7 and 2. Agreement 50 and 24. Cosine 1.00 and 0.96.
- Remember: a score only ranks. Read the top three. Meaning search finds descriptions, exact search finds codes.
- Ask the vendor: which measure, and were the vectors set to length 1 first?

I keep: The distance cheat sheet: four ways to say how close two things are, each with its picture, its rule and when to use it.

Session 5

Language models: tokens, attention and next-word probability

How does a language model turn your words into an answer, and why can the same question give different answers?

- Language model on a page
- Loop: text so far, score every piece, pick one, add it, repeat.
- Token: a piece of text. Numbers, names and other languages split into more pieces.
- Softmax: raise 2.72 to each score, divide each by the total. Shares add to 100.
- Temperature: divide scores first. Low repeats, high wanders.
- Attention: which earlier words matter to this word. One head shows one pattern.
- Five questions for any answer: Where did this come from? Can I check it against a source? Would it change if I asked again? Does it contain figures or quotes? Who reviews it?
- Recorded runs: GPT-2 small, 124M parameters, 6 November 2026.

I keep: A one-page sheet: the next-word loop, the softmax recipe, what temperature does and five questions to ask about any language-model answer.

Session 6

Foundation models: pretraining, post-training, model families, reasoning and multimodal models, benchmarks

What is a foundation model, how does it become an assistant, and what does a benchmark score tell me?

- Model comparison grid
- My task (one line): _____
- Models to try (2 or 3): _____
- Ten real cases with the answer I expect: attach a list
- Three checks (for example figures, label, nothing invented): _____
- Model card questions: licence? size? trained on? limits? what is missing?
- Benchmark claim questions: which test? what setup? what date? how does it match my task?
- Recorded runs: 6 November 2026. Qwen2.5-1.5B-Instruct and GPT-2 run locally; Claude in a fresh subagent session. Not ChatGPT or Gemini.

I keep: A one-page model comparison grid for your own tasks, with the questions to ask of a model card and a benchmark claim.

Session 7

Open weights, file formats, size and running cost

What does it mean to own, host or rent a model?

- RENT, HOST OR OWN: DECISION SHEET
- Task: _____ Data it touches: _____ Sensitive? (yes or no)
- Memory sum: parameters x bytes per number = size. Example: 7 billion x 2 bytes = 14 GB; 7 billion x 0.5 byte = 3.5 GB
- Open source or open weights? Which licence? Which uses does it limit?
- Container (Safetensors, GGUF, ONNX)? Bits per number (16, 8, 4)? Quality tested on my cases?
- Route: rent / host / own / wait. Reason: _____
- One thing that would change my mind: _____

I keep: A one-page rent, host or own decision sheet, with the memory sum and the licence questions to ask.

Session 8

Prompting and reasoning models

How do I get reliable work out of an assistant?

- PROMPT TEMPLATE AND LOG
- Five parts: Role. Context. Examples. Format. Checks (and what to say when unsure).
- My prompt: _____
- Test cases with known answers (three to ten): _____
- Log row: Version ___ What changed ___ Runs ___ How many right ___ Next change ___
- Who checks the figures before use: _____
- Template: Templates/Reusable Brief Template.docx (seven parts)

I keep: A reusable prompt template and a prompt log, so that every prompt you keep has been tested.

Session 9

Giving a model your knowledge: retrieval-augmented generation, long context and fine-tuning

How do I give an AI model my company's knowledge, and when is each way worth it?

- PROMPT, RETRIEVE OR FINE-TUNE: ask in this order.
- 1. Does a clearer prompt with the page pasted in give right answers on ten checked questions? Yes: stop.
- 2. Are the facts too many, too long or changing? Retrieve. Check your chunks. Let the model say not stated.
- 3. Is the style or format still wrong after steps 1 and 2? Consider fine-tuning, with examples you own and personal data removed.
- Always: measure on the same ten questions before you climb a rung.
- Remember: a high similarity score shows that topics are close. It does not show that the answer is present. Fine-tuning shapes style and does not reliably teach facts.
- Worksheet: six NVF requests. (a) Summarise one supplier email. (b) Answer staff questions from a handbook that changes each quarter. (c) Every customer reply ends with the house sign-off. (d) Check a price against one pasted quote. (e) Answer questions about 400 contracts. (f) Draft replies in the Sales Manager voice from 200 past emails.

I keep: A one-page decision tree: prompt first, then retrieve, then fine-tune, and the test that tells you when to climb.

Session 10

When models fail, and how to check them

How do I find out when an AI answer is wrong, before it costs me?

- FIVE-STEP CHECK. The more an error would cost, the more steps you take.
- 1. Source: ask for the quote or clause, then open it.
- 2. Recount: redo every number by hand or with a formula.
- 3. Compare: check the key, or a second source.
- 4. Run twice: if the same request gives different answers, look closer.
- 5. Sign: a named person decides and records it.
- TEST-SET TEMPLATE. Id | Question | Known answer | Source | Run 1 | Run 2 | Cost of a wrong answer. Ten questions, two with no answer. Score right, missed, invented. Compare with the baseline of always saying not stated. Run again after any change.
- Recorded finding: most runs were right and the faults were small. Only checking found them.

I keep: A five-step check routine and a test-set template, each on one page.

Session 11

Agentic AI: tools, loops, MCP, autonomy and guardrails

What changes when an assistant can act as well as answer?

- Tool Boundary Sheet: fill it before any agent is switched on.
- The loop: plan, act, observe, repeat; it stops when the goal is met, a step limit is reached or a person is needed.
- Badges: give the smallest set that does the job. Read and draft are enough for a mail summary.
- Four rows: Read, Write, Send, Spend. For each: it may, a person approves, it must never.
- The gate: sends, payments, deletions and commitments always need a named person.
- Controls: a log line for every step, one owner, stop conditions, an off switch that works in minutes.
- Planted instructions: addressed to the AI, ask for data to leave or a deletion, ask for silence. Stop and tell the owner.
- MCP: a standard plug (host, one client per server; servers offer tools, resources, prompts). It connects; it does not decide permissions.
- Levels: 0 answers, 1 suggests, 2 drafts and you approve, 3 acts then reports, 4 acts alone. Choose the lowest that works.

I keep: Your Tool Boundary Sheet: what an agent may read, write, send and spend, who approves, the log and the off switch.

Session 12

Workflow mapping (practical)

Which of my tasks should be a rule, an AI step or an agent?

- Workflow map: trigger, steps, handoffs (red), check (gold), one lane per person or system.
- Decide each step in order: clear words or numbers: rule. Meaning with a quick check: AI step. Several tools and its own next step: agent with a boundary. Money, trust or law: person.
- Side by side on twelve customer messages: rule 7 of 12; AI 9 exact, 11 acceptable, 1 wrong with confidence (M14). Different measures; small sample.
- Hours saved = times a month x people x minutes saved each time / 60. Review time is a cost.
- NVF plan example: 4 x 80 minutes = 5.3 hours a month; net UGX 159,000 a month; payback about 3 months (invented inputs).
- Assistant sketch: trigger, inputs, outputs, fewest badges, the gate, the off switch.
- Measure four weeks: minutes each run with review, errors and their cost, cases sent to a person, your stop rule.
- Templates: Task Card, Use-Case Scorecard, Tool Boundary Sheet, Business Case Workbook (Inputs sheet).

I keep: Your mapped workflow, the choice for each step, your assistant sketch and the hours-saved estimate in the Business Case Workbook.

Session 13

Governance and buying (practical)

What rules must my company set for AI, and how do I question what I am sold?

- AI use policy: eleven sections. I drafted 3 approved tools, 4 data rules, 5 human review, 7 accounts, 8 incidents, 10 review date.
- Four data questions: living person? harmful if leaked? staff only? already published? The strictest answer wins.
- Risk score = likelihood (1 to 5) x impact (1 to 5). High 15 or more, Medium 8 to 14, Low below 8.
- Every limit of a model gives a rule: checker, approval on every send, data classes, licence read, narrow access, re-test after change.
- Vendor red flags: no number, no approval, trains on my data, no references, no cap, no exit, a deadline, bank login.
- Five steps: ask in writing, score 0 to 2, ask for proof, pilot with a stop rule, sign with an exit.
- Law, my understanding: the Data Protection and Privacy Act 2019 covers personal data (sections 19, 20, 21, 23, 29). Ask my lawyer.
- Frameworks: NIST AI RMF, ISO/IEC 42 001, OECD AI Principles, African Union Continental AI Strategy, EU AI Act (only if I serve the EU).

I keep: Your Risk Register, your AI Use Policy draft, your Vendor Question List with five questions, and your page of questions for the lawyer.

Session 14

Data readiness assessment, roadmap and 90-day plan (practical)

How ready is my organisation's data for AI, what will I fix first, and who will own it?

- Five kinds of data: tables, text, images, sound, sensors. First steps: define columns, find the current version, read text out of images, transcribe sound, check gaps.
- Readiness scorecard: Inventory, Readable, Complete and unique, Metadata, Rights, Current. Green 2, Amber 1, Red 0. Total out of 12. A Red on rights stops the work.
- Six steps: inventory, digitise, clean, describe, protect, index and test. Start with one folder and one task.
- First test: 20 questions with known answers, two or three traps, a line for success written first.
- Every roadmap action: owner, start, due date, hours, depends on, a success measure someone else can check.
- Check-ins: day 30 (controls in place? gap 1 closed?), day 60 (stop rule met? actions on time?), day 90 (re-score, then continue, expand, pause or stop).
- Within 24 hours: book the three dates in my diary and tell each owner their action.
- The follow-up kit: 30-60-90 Check-in Form, Results Tracker, Sponsor Briefing.

I keep: Your inventory and scorecard, your roadmap with its risks and decisions, and the Follow-up Kit: Check-in Form, Results Tracker and Sponsor Briefing.